	0 h
2(a)	The atoms of ideal gas are in continuous random motion and have equal probability of moving in any direction. Hence there will be just as many atoms moving in one direction as that in the opposite direction. The mean velocity of the atoms in any direction will be equal and opposite to their mean velocity in opposite direction. Since velocity is a vector quantity, these mean velocities will cancel out and the mean velocity of the atoms of the ideal gas will be zero.
(b)	Using ideal gas equation, $pV = nRT = NkT$

	$\Rightarrow N = \dfrac{pV}{kT}$ $= \dfrac{3.6 \times 10^{5} \times 4.2 \times 10^{-3}}{1.38 \times 10^{-23} \times (273 + 70)}$ $= 3.193 \times 10^{23}$ $= 3.2 \times 10^{23}$ (2 sig. fig) Hence the sample of gas contains 3.2×10^{23} atoms
(c)	Volume of the gas atoms = $3.2 \times 10^{23} \times \dfrac{4}{3}\pi\left(\dfrac{d}{2}\right)^{3}$ $= 3.2 \times 10^{23} \times \dfrac{4}{3}\pi\left(\dfrac{2 \times 10^{-10}}{2}\right)^{3}$ $= 1.3 \times 10^{-6}$ m^3
(d)	$\dfrac{\text{Volume of gas atoms}}{\text{Volume occupied by gas}} = \dfrac{1.3 \times 10^{-6}}{4.2 \times 10^{-3}} = 3.1 \times 10^{-4}$ Since the ratio of volume of gas atoms to volume occupied by gas is 3.1×10^{-4}, it implies that the volume of the gas atoms is negligible compared to the volume occupied by the gas, thus providing evidence that the gas is ideal.
3(a)(i)	t and t